A Setting of τ in the relation index.

The setting of the threshold $\tau = 2.2$ for the value of $\frac{A^h_{j,s}}{\max_{k \neq s}\{A^h_{j,k}\}}$ is based on our observation in the distribution of values of $\frac{A^h_{j,s}}{\max_{k \neq s}\{A^h_{j,k}\}}$. Using input sentences and corresponding triplets in the AGENDA test set, we computed the value of $\frac{A^h_{j,s}}{\max_{k \neq s}\{A^h_{j,k}\}}$ at all heads h and all current tokens t_j that satisfy $s = \arg\max_k\{A^h_{j,k}\}$. The distribution of this value is shown in Figure 11. The frequency of values larger than 2.2 dropped significantly to less than 5%. Thus, we empirically set the threshold $\tau = 2.2$ to only focus on attention heads that *exclusively attended to the head token*.

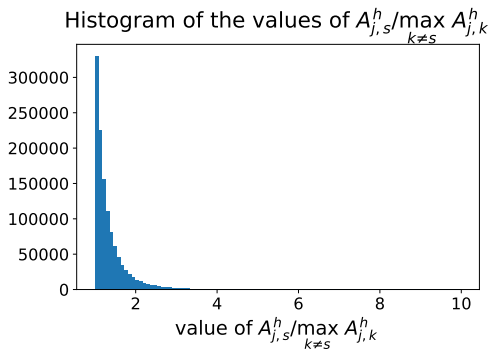

Figure 11: Distribution of $\frac{A^h_{j,s}}{\max_{k \neq s}\{A^h_{j,k}\}}$ in InternLM2-1.8B. For better visualization, we just show the distribution of $\frac{A^h_{j,s}}{\max_{k \neq s}\{A^h_{j,k}\}}$ within the range of 0 10.

Moreover, we also conduct ablation experiments with different values of τ. Specifically, we compute the relation index for syntactic dependencies on InternLM2-1.8B with a smaller value $\tau = 2.0$ and a larger value $\tau = 2.5$, respectively. Heatmaps in Figure 15 show that the setting of τ does not significantly affect the distribution of the relation index. Different settings of τ yield a similar set of semantic induction heads that have a high relation index.

B Relation index for the reverse semantic relationships

The semantic relationships in knowledge graphs are bidirectional, thus we also compute the relation index of attention heads for the reverse semantic relationships. Figure 16 shows the results in InternLM2-1.8B. Comparing Figure 16 and Figure 4, we find that some attention heads represent both directions of the relation.

C Format compliance ability

Besides the format accuracy in Figure 6, we also measure the format compliance ability from another perspective: the minimum number of shots required to achieve a format accuracy of over 80%. A lower minimum number of shots indicates a better format compliance ability. We set a maximum limit of 20 shots. If the model fails to achieve an accuracy of 80% even with 20 shots, we record the result as 20. Figure 12 consistently shows that the model learns format compliance on simple tasks earlier than on complex tasks, but all achieve a good performance at 1000 steps.

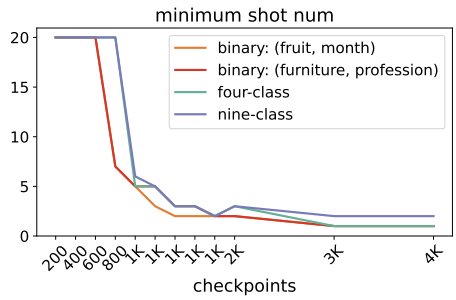

Figure 12: The minimum number of shots (examples in the prompt) required to achieve over 80% format accuracy.

Moreover, considering that the format of generating "true" or "false" in relation justification tasks is different from the simpler format in classification tasks in Figure 6, we additionally examine the format accuracy on relation justification tasks in Figure 17. The format compliance in these tasks emerges from about 2K steps, later than that in simpler tasks in Figure 6, and achieves about 50%-70% at 15K steps.

D Pattern discovery tasks are not binary classification tasks

Unlike binary classification tasks, pattern discovery tasks are actually more difficult for generative models.

First, the model generates the next token as the prediction, which is different from the classic binary task. When generating the next token, there are a total of $|\mathcal{V}| = 92544$ candidates in the vocabulary $\mathcal{V}$. If the model could perfectly comply with the format, the problem is simplified as a binary classification task. However, models in the early stages of training do not have such ideal capabilities yet. For example, Figure 17 shows that the format compliance in relation justification tasks

emerges from about 2K steps, later than that in simpler tasks in Figure 6, and achieves about 50%-70% at 15K steps. Thus, the exact prediction accuracy of the model will be lower. Second, the model might inherit certain biases from the training data. Thus, the probability of generating either the token "true" or "false" is not 0.5 vs 0.5.

E Pattern discovery ability of a well-trained model

Although we have observed the development of the pattern discovery ability in Figure 7, we find that it is hard to be fully mastered by the model. Figure 13 reports the prediction accuracy of the well-trained InternLM2-1.8B on classification tasks. Even with 100 examples in the prompt, the model still cannot perfectly recognize and utilize the pattern in the prompt. On the other hand, Figure 14 shows that InternLM2-1.8B achieves higher accuracy on relation justification tasks.

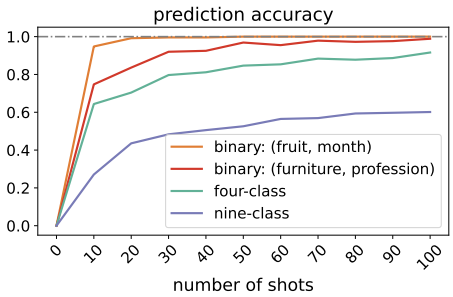

Figure 13: Prediction accuracy of InternLM2-1.8B on classification tasks.

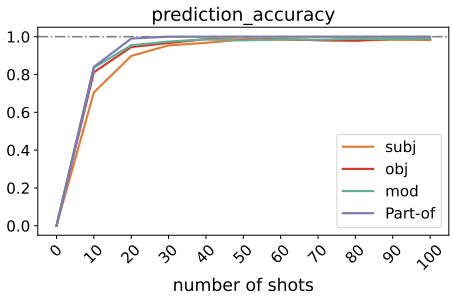

Figure 14: Prediction accuracy of InternLM2-1.8B on relation justification tasks.

F Change of relation index for semantic relationships over training time

This section provides results of the change of relation indexes of attention heads for semantic relationships in knowledge graphs. We consider both directions of the semantic relationship, and results are shown in Figure 18 and Figure 19.

G Grouping attention heads for relations

As discussed in Section 5, the semantic relationships are dependent on the context, so they may be stored in different attention heads given different contexts. In this section, we perform the grouping analysis on semantic relations in knowledge graphs.

Figure 20 and Figure 21 show the occurrence times of each attention head having the largest value of $\mathbb{E}_j[a_T^{h,j}]$ for triplets in semantic relations and the reverse triplets. We observe that some attention heads are commonly highlighted in different types of relationships.

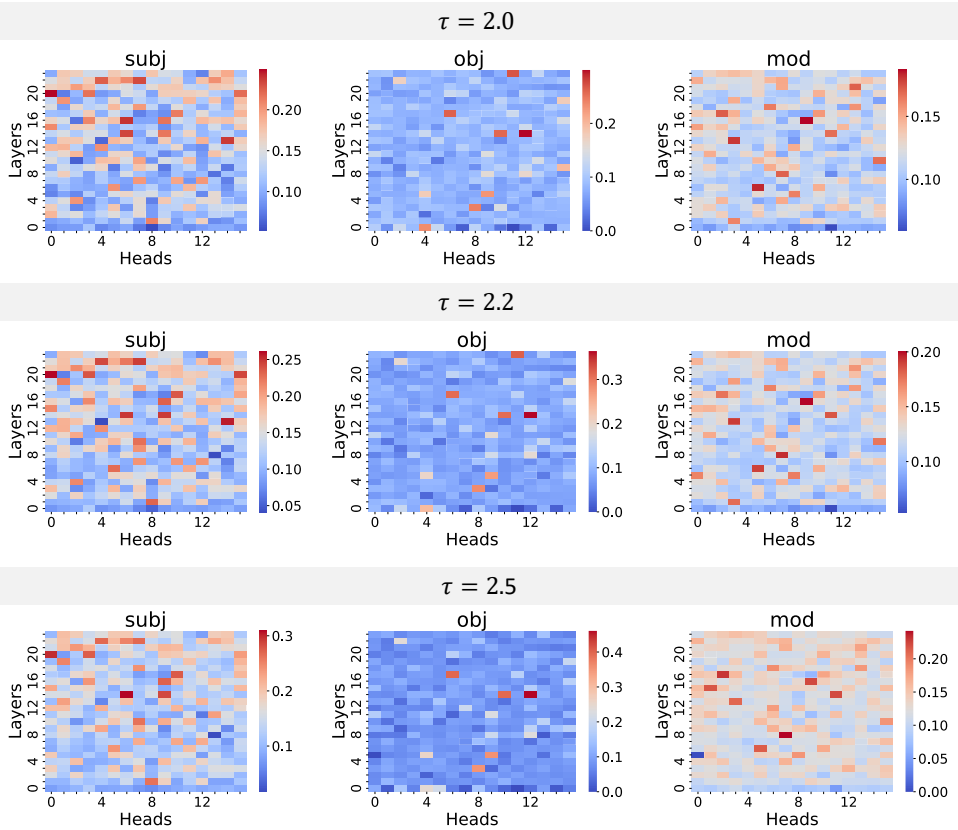

Figure 15: Heatmaps of relation indexes with different settings of τ. Different settings of τ yield a similar set of semantic induction heads that have a high relation index.

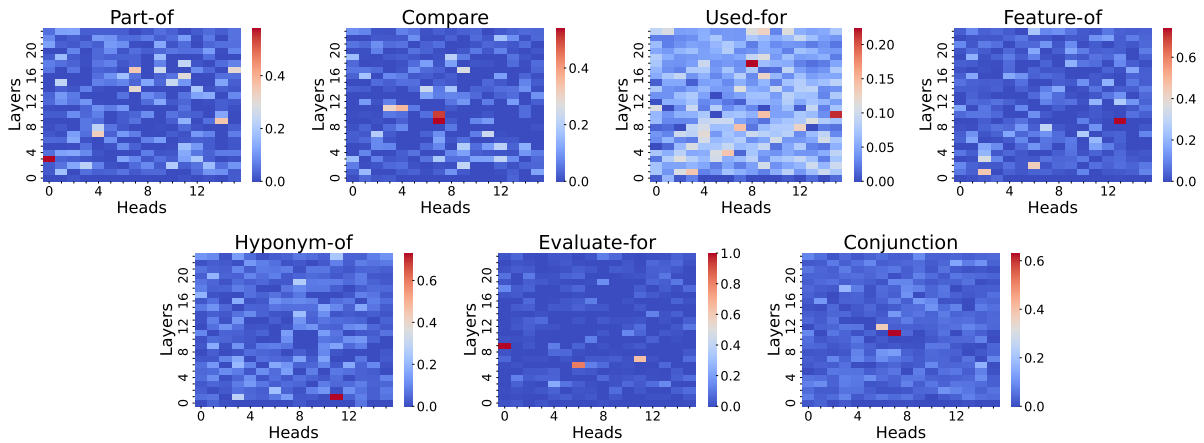

Figure 16: Heatmaps of the average relation index of attention heads for the reverse semantic relationships in knowledge graphs.

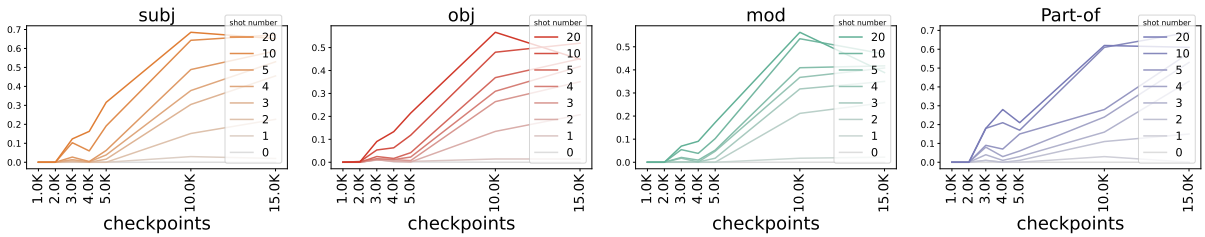

Figure 17: Format accuracy of different relation justification tasks at different checkpoints. Each line represents the format accuracy with different numbers of shots (examples) in the prompt.